Hybrid MES-Knapsack Allocation for Fair and Efficient Participatory Budgeting

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***Abstract*—Participatory Budgeting (PB) is an in democratic method which allows citizens to directly vote on how public funds are allocated among different community projects. As the processes grow in size and complexity, the effective and equitable allocation procedures become increasingly important. This paper examines and analyzes four algorithmic methods for participatory budget allocation: the Greedy strategy, the Knapsack optimization algorithm, the Method of Equal Shares (MES), and a suggested Hybrid MES-Knapsack model. Greedy algorithm provides efficient and easy heuristics but is not fair and optimal for complex preference distributions. The Knapsack model optimizes utility maximization but is not proportional in representation or fair to voters. MES provides fairness and proportionality through equal budget shares per voter but usually leaves non-exhaustive budgets and is computationally expensive. In an effort to alleviate these disadvantages, we present a Hybrid MES-Knapsack algorithm that maintains MES's fairness promises but maximizes fund usage through a Knapsack-based optimization.**

***Index Terms*—Participatory Budgeting, Resource Allocation, Method of Equal Shares (MES), Knapsack Problem, Greedy Algorithm, Hybrid Algorithm, Budget Optimization, Fairness in Voting, Public Budget Distribution**

I. Introduction

Participatory Budgeting (PB) is democratic method that lets citizens directly determine the use of public funds. Since its emergence in Porto Alegre, Brazil, in the late 1980s, PB has become recognized internationally as a powerful civic engagement, transparency, and community-led development tool. The process is often characterized by gathering project proposals, having citizens vote on them, and distributing the available budget according to people's preferences. Yet, as PB extends to larger groups and more complicated sets of projects, the necessity for computational approaches to distribute funds efficiently, justly, and comprehensively grows more apparent. Manual methods, which have served in the past, are not feasible at scale, and algorithmic solutions are therefore being developed to mechanize the budget distribution without sacrificing fairness and proportionality. Of these algorithmic approaches, the Greedy method is frequently considered on account of its simplicity and computational efficiency. efficiency. It chooses projects on the basis of the highest cost-benefit ratio or number of approvals, but often ignores voter equity and diversity in project choice. In contrast, the Knapsack algorithm represents budget allocation as an optimization problem, seeking to maximize the sum of utility within the available budget. Although more optimal in utility, it does not necessarily guarantee fair representation of diverse voter preferences. To overcome these limitations, the Method of Equal Shares (MES) has been proposed within the computational social choice literature. MES distributes the budget proportionately across voters and chooses projects that can be financed equitably from these equal portions. MES ensures proportionality and fairness but generally lacks completeness of the budget and can cause underallocation of available money. In this paper, we suggest a Hybrid MES-Knapsack method, integrating MES fairness with the efficiency of the Knapsack model. Following an initial assignment after applying MES for proportionality, the leftover budget is optimized again using a Knapsack-based mechanism, seeking to maximize utility while not sacrificing fairness. This hybrid model attempts aiming to achieve a balance among fairness, efficiency, and full budget usage, and is therefore an attractive solution for large-scale participatory budgeting scenario.

II. Literature Survey

In participatory budgeting, different algorithmic techniques have been developed to find a balance among fairness, computation, and usage of the budget. This section summarizes four main methods—Greedy, Knapsack Optimization, Method of Equal Shares (MES), and Hybrid Approaches—describing their mechanisms, advantages and limitations.

A. Greedy Approach

Mechanism: The Greedy strategy works by picking projects in a descending order of a utility function, typically the votes-per-cost ratio. The algorithm proceeds to choose projects so long as the rest of the budget is enough to pay for the next project from the ordered list.

Advantages: This algorithm is simple and computationally fast. It is simple to carry out and works well with larger numbers of projects and voters. Moreover, it tends to maximize the use of a large proportion of the budget by ranking projects providing the highest value per unit expenditure.Additionally, it often ensures that projects offering the highest value per unit cost are prioritized, leading to effective utilization of a significant portion of the budget.

Limitations: Even though the Greedy approach is efficient, it has a tendency to give preference to projects endorsed by the majority of voters at the expense of those endorsed by minority or underrepresented groups. This may lead to unequal results and decreased diversity in chosen projects.

B. Knapsack Optimization

Mechanism: Knapsack-based approaches pose the budget allocation issue as a 0/1 Knapsack problem. Each project is viewed as an item with a cost (weight) and a utility value (typically vote totals). The objective is to select a subset of projects maximizing the aggregate votes while remaining within the given budget.

Advantages: The Knapsack model yields a mathematically optimal solution in terms of vote maximization. Ensure that the chosen projects collectively receive the maximum number of votes possible within the budget constraints provided, thus maximizing efficiency.

Limitations: Nevertheless, this utility-maximizing strategy does not consider the distribution of the voter's interests. Therefore, projects that are important to minority groups but do not have broad support may be omitted, resulting in outcomes that might not represent fair representation.

C. Method of Equal Shares (MES)

Mechanism: MES tries to guarantee fairness by distributing the entire budget proportionally among all voters. A project is being funded only if the set of its supporters together can pay for it with their equal portions of the budget. This mechanism guarantees that every voter's power is equal when making an allocation decision.

Advantages: The first strength of MES is that it is committed to proportional fairness. It safeguards the interests of majorities and minorities by permitting any sufficiently supported project to be financed as long as its supporters are able to pool their equal shares.

Limitations: A well-known limitation of MES is that it has a tendency to under-utilize the available budget. In practice, only 60-80of the stringent affordability condition, which could result in prevention of funding of otherwise worthwhile projects if the aggregate budget of their supporters is insufficient.

D. Hybrid Approaches

Recent research has investigated the synergy of the best of MES and optimization approaches to maximize fairness and budget utilization. The hybrid approaches identify the trade-off between maximizing the two objectives:

Pure MES solutions, being fair, are often inefficient in that they leave significant parts of the budget unused. Pure Knapsack models, while efficient in maximizing utility, are not able to provide fair representation of minority groups.

Our Hybrid Approach remedies this trade-off by combining both approaches in a two-stage process:

1) Phase 1: Utilize the MES mechanism to make the initial allocation, thus ensuring proportional fairness and providing each voter with equal influence over some portion of the budget.
2) Phase 2: Use Knapsack optimization on the remaining budget, aiming to fund additional projects that maximize utility without pre-empting the fairness of the initial allocation.

This method tries to find a pragmatic compromise, maintaining fairness while ensuring a more comprehensive and effective utilization of public funds.

III. Methodology

The approach operates in two phases: an initial allocation using the Method of Equal Shares (MES) and a refinement phase using Knapsack optimization to utilize leftover budget effectively.

A. Hybrid Knapsack-MES Algorithm

Input:

- A set of projects $P = \{p_1, p_2, ..., p_m\}$, each with associated cost $c(p_i)$.
- Voter approval ballots $A_1, A_2, ..., A_n$, where each $A_j \subseteq P$.
- Total available budget B.

[htbp] Hybrid MES + Knapsack Algorithm [1] **Input:** Project set P, Budget B, Voter preferences V $W \leftarrow \text{MES}(P, V, B)$ Initial MES allocation $B' \leftarrow B - \text{cost}(W)$ Remaining budget $U \leftarrow P \setminus W$ Unselected projects $W' \leftarrow \text{Knapsack}(U, V, B')$ Greedy or exact Knapsack **return** $W \cup W'$ Final funded project set

Phase 1: MES Allocation

1) Distribute the total budget equally among all n voters:
$$b_i = \frac{B}{n}, \quad \forall i \in \{1, ..., n\}$$
2) Iteratively fund projects where the total contribution from their supporting voters meets or exceeds the project's cost:
$$\sum_{j:p \in A_j} b_j \geq c(p)$$
3) Deduct the corresponding amount from the virtual budgets of supporting voters for each funded project.
4) Terminate MES phase when no more projects are affordable under the current budget distribution.

Phase 2: Knapsack Optimization

1) Identify underrepresented voters, i.e., voters who retain more than 50% of their original virtual budget after the MES phase.
2) For each remaining unfunded project, compute a fairness-adjusted score:
$$\text{Score} = \frac{\text{Support from Underrepresented Voters}}{\text{Cost}} \times \text{Cohesion}$$
Where:
   - Support from Underrepresented Voters = number of approvals from underrepresented voters.
   - Cohesion is a measure of agreement or intensity of support (e.g., ratio of underrepresented to total approvals).
3) Select projects using a Knapsack optimization algorithm that maximizes the total score while remaining within the unused portion of the budget.

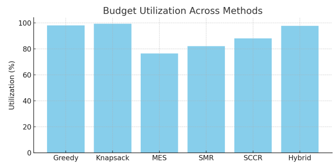

Fig. 1. Budget Utilization

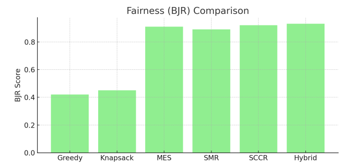

Fig. 2. Fairness

Method	Fairness	Voter Simplicity	Budget Utilization	Minority Representation
Greedy	Low	High	High	Low
Knapsack	Low–Medium	Medium	High	Low
MES	High	Medium	Low–Medium	High
Hybrid (MES+Knapsack)	High	Medium	High	High

Fig. 3. Comparision

Output:

- Final set of funded projects.
- Total budget utilization.
- Fairness indicators and representation metrics.

B. Comparison Metrics

- **Budget Utilization:**

$$\text{Budget Utilization} = \frac{\text{Total Cost of Funded Projects}}{B} \times 100$$

- **Fairness (Gini Coefficient):** Measures inequality in voter satisfaction.
- **Minority Representation:** The proportion of underrepresented voters who have at least one approved project funded.

IV. Results and Discussion

For analyzing the efficiency of the suggested Hybrid Knapsack-MES methodology, we compared its performance against four key measures: fairness, voter simplicity, budget allocation, and minority representation. The methodologies considered are the Greedy approach, Knapsack optimization, Method of Equal Shares (MES), and our hybrid model.

The MES and Hybrid methodologies attain the best levels of fairness by their own nature. MES provides each voter an equal portion of the budget, tending to include proportional representation. The Hybrid model preserves this fairness in its first phase and reinforces it with underrepresented-aware optimization during the second phase. Greedy and Knapsack methods, on the other hand, have lower fairness, tending to ignore minority preferences to favor majority-backed, high-utility projects.

Voter Simplicity: The Greedy method has the highest score in voter simplicity, since it only considers approval counts and is very easy to interpret. MES and Hybrid need a more complex understanding of budget allocation, yet still remain capable to voters. Knapsack adds complexity to utility estimation, thus making it less clear to the general participant.

Budget Utilization: Both Greedy and Knapsack methods exhibit high budget utilization owing to their efficiency-driven selection principles. MES, however, tends to leave a considerable amount of the budget unused due to its requirement. The Hybrid model mitigates this drawback by employing MES for equitable allocation and Knapsack for expending remaining budget, thereby satisfying both equity and efficiency.

Minority Representation: MES and Hybrid models unequivocally promote minority representation by financing projects even if backed by smaller, united voter constituencies. The Greedy and Knapsack approaches, however, tend to ignore such constituencies because of their majority-oriented reasoning.Hybrid approach is better than MES because it guarantees that even underrepresented voters make a significant input into the end allocation,particularly in the Knapsack stage.

V. Conclusion

In this paper, we suggested and tested a Hybrid Knapsack-MES algorithm for participatory budget distribution, seeking to balance the usually competing demands of fairness, efficiency, and inclusiveness. Through the union of the proportional fairness of the Method of Equal Shares (MES) and the optimization power of the Knapsack algorithm, our solution remedies main limitations of current methods.

The MES step guarantees that all voters have an equal say in a fraction of the budget, facilitating equitable representation and minority inclusion. The following Knapsack phase information- rationally distributes the available funds by favoring projects supported by underrepresented voters, thus increasing both budget utilization and equity. Comparative analysis reveals that the hybrid model per- consistently dominates single-approach alternatives across several criteria—attaining high fairness, robust minority representation, and close-to-optimal budget utilization without incurring large complexity. These results highlight the power of hybrid mechanisms in real-world participatory budgeting contexts particularly where democratic representation and resource efficiency are both important.he MES phase ensures that every voter has equal influence over a portion of the budget, supporting fair representation and minority inclusion. This work provides a foundation for future research, such as real-time applications, dynamic budget resizing, and extensions to ranked or weighted vote. In general, the Hybrid Knapsack-MES algorithm is a realistic and fair solution for deliberative governance in large and heterogeneous populations.

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